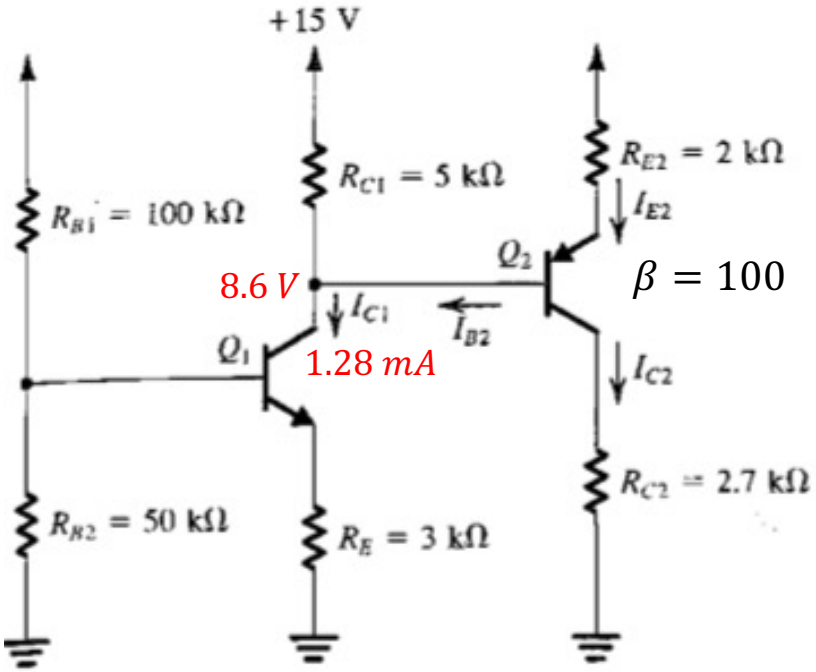
Bipolar Junction Transistors

- DC Analysis (Biasing) of BJTs. Saturation Mode.
- IV and Transfer Function curves.

Examples

9) Another transistor stage is added to example (4).



$\beta = 100$

$$V_{B2} = V_{C1} = 8.6\ V$$

$$V_{E2} = 8.6 + 0.7 = 9.3\ V$$

$$I_{E2} = \frac{15 - 9.3}{2\ k} = 2.85\ mA$$

$$I_{C2} = 2.82\ mA$$

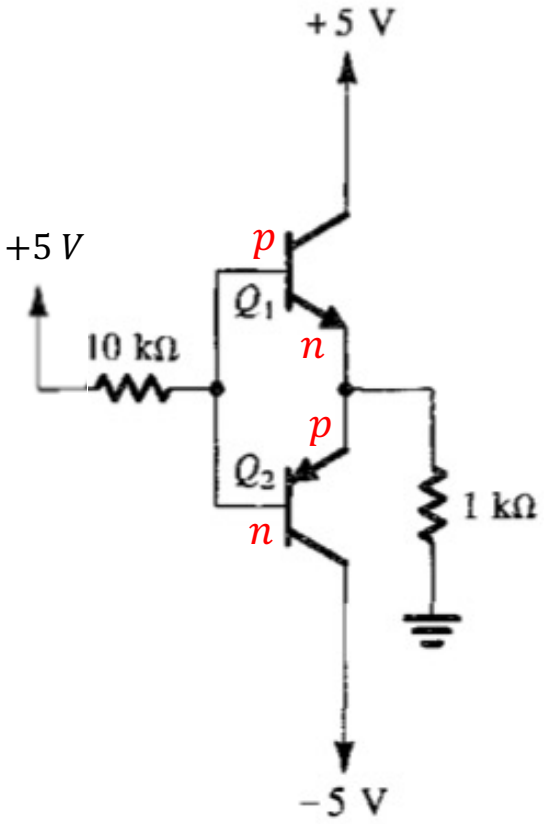
$$V_{C2} = (2.82\ m)(2.7\ k) = 7.62\ V$$

Active mode

$$I_{B2} = 28.2\ \mu A$$

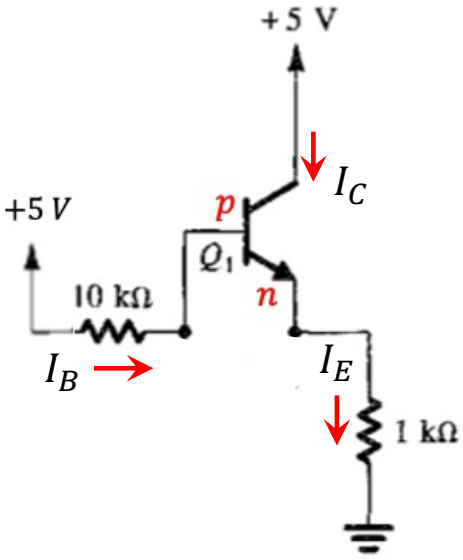
Examples

10) Analyze the DC BJT circuit below. (β=100).



The two transistors cannot be both ON, because the EB junctions are connected across each other.

With the given biasing, Q_2 is OFF.



Assume active mode

$$V_{BE} = 0.7\ V$$

KVL in the B-E circuit

$$5 - I_B(10\ k) - 0.7 - (\beta + 1)I_B(1\ k) = 0$$

$$I_B = 38.7\ \mu A$$

$$I_C = 3.87\ mA$$

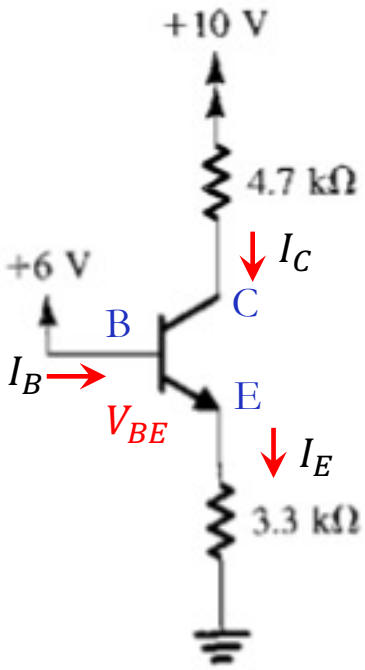
$$I_E = 3.91\ mA$$

$$V_B = 5 - (38.7\ \mu)(10\ k) = 4.6\ V < V_C$$

Active mode

Examples: Saturation Mode

11) Analyze the DC BJT circuit below. (β=100).



Circuit was checked in a previous example. It was not in active mode.

$$V_{BE} = 0.7\ V = V_B - V_E$$

$$V_B = 6\ V$$

$$V_E = 6 - 0.7 = 5.3\ V$$

$$I_E = 1.61\ mA$$

The BC junction is also forward. Then,

$$V_{BC} = 0.5\ V = V_B - V_C$$

$$V_C = 5.5\ V$$

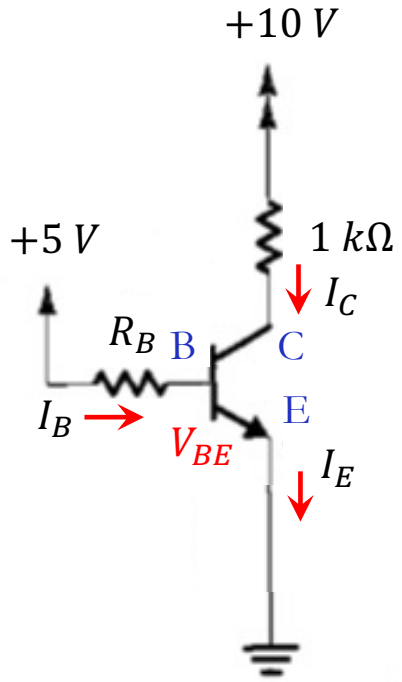
$$I_C = \frac{10 - 5.5}{4.7\ k} = 0.96\ mA$$

$$I_B = I_E - I_C = (1.61 - 0.96) = 0.65\ mA$$

$$\beta_{forced} = I_C/I_B = 1.48$$

Examples: Saturation Mode

12) Design the circuit for deep saturation with overdrive factor of 10. (β=50-100).



Saturation means both junctions are forward

$$V_E = 0\ V$$

$$V_B = 0.7\ V$$

$$V_C = 0.2\ V$$

Use the minimum β for the design.

$$I_C = \frac{10 - 0.2}{1\ k} = 9.8\ mA$$

$$I_B = \frac{I_C}{\beta_{min}} = \frac{9.8\ m}{50} = 0.196\ mA$$

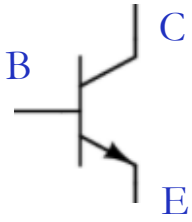
$$I_{B_sat} = 10 \times 0.196\ m = 1.96\ mA$$

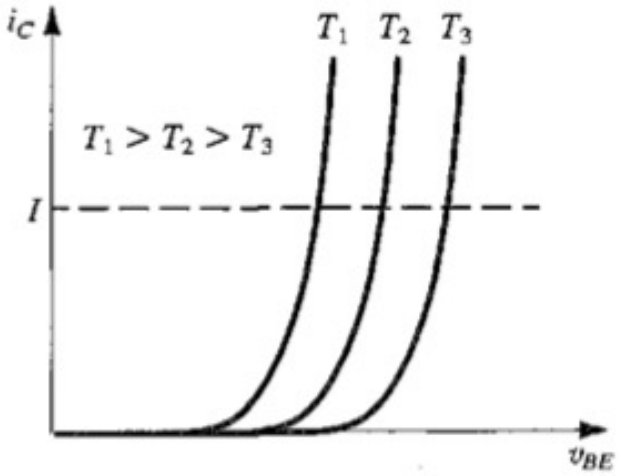
$$R_B = \frac{5 - 0.7}{1.96\ m} = 2.2\ k\Omega$$

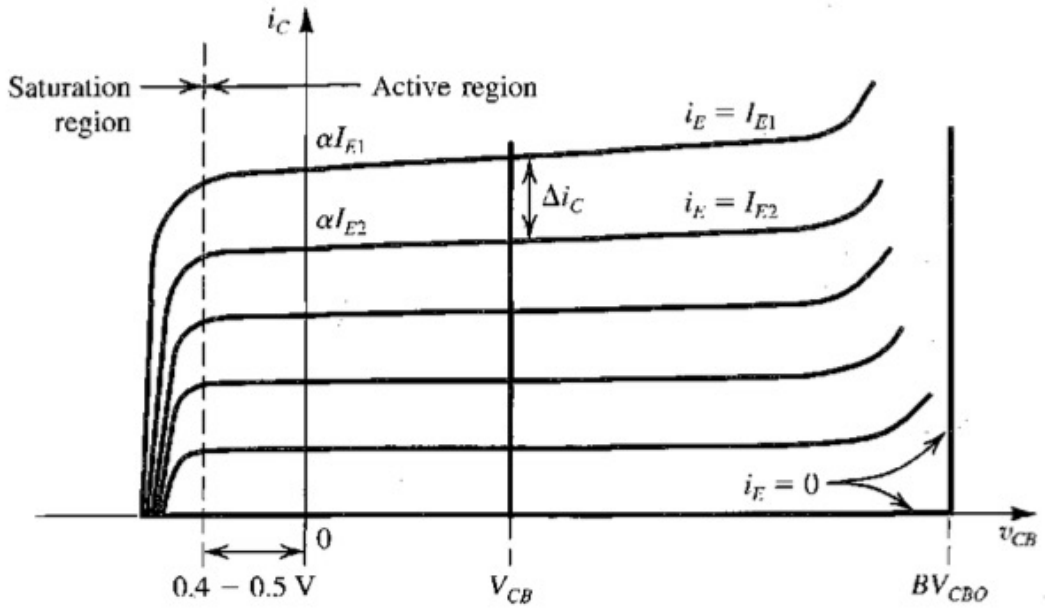
IV Characteristics

The current in the collector is a result of a forward-biased pn junction (EB junction)

$$I_C \approx I_s \, e^{V_{BE}/V_T}$$

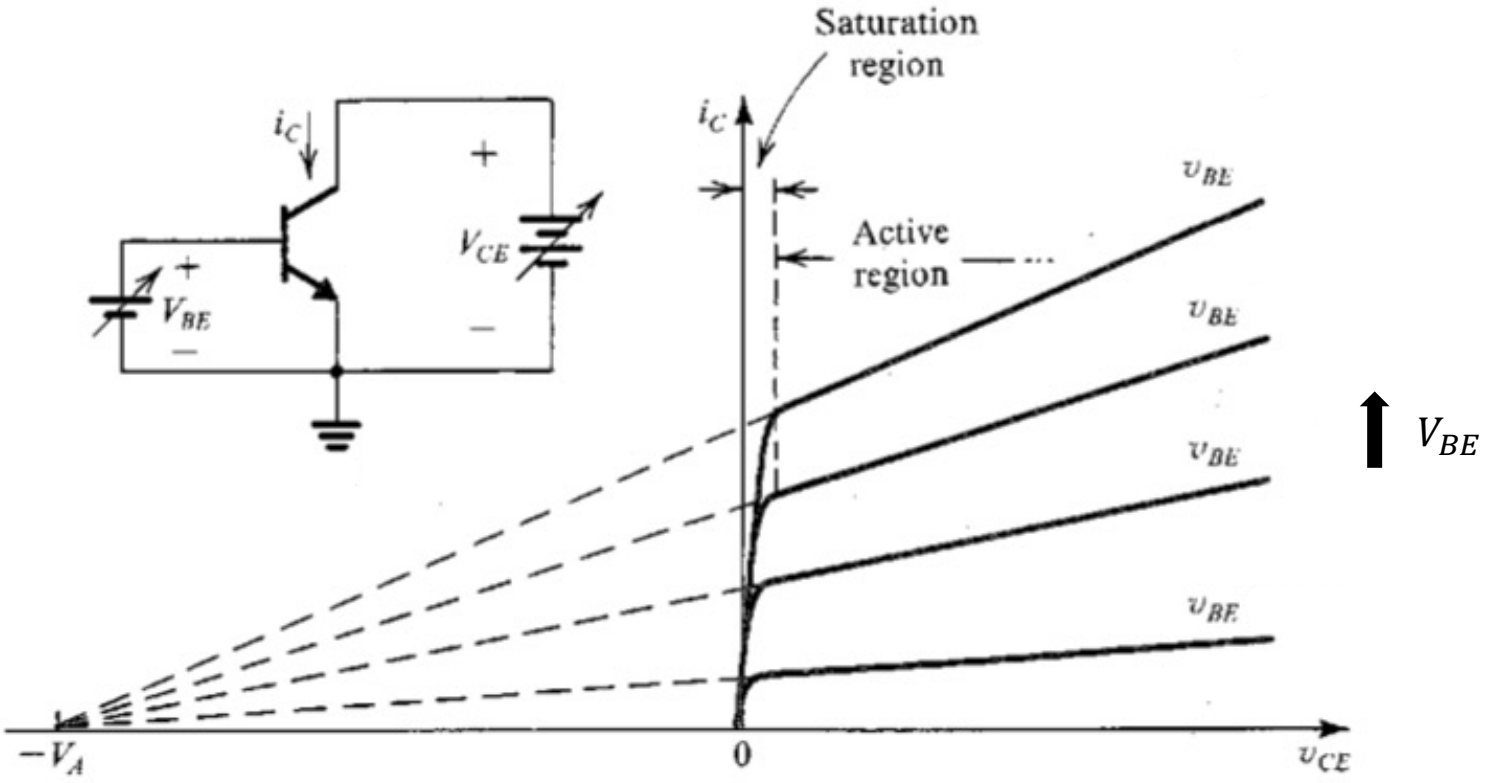






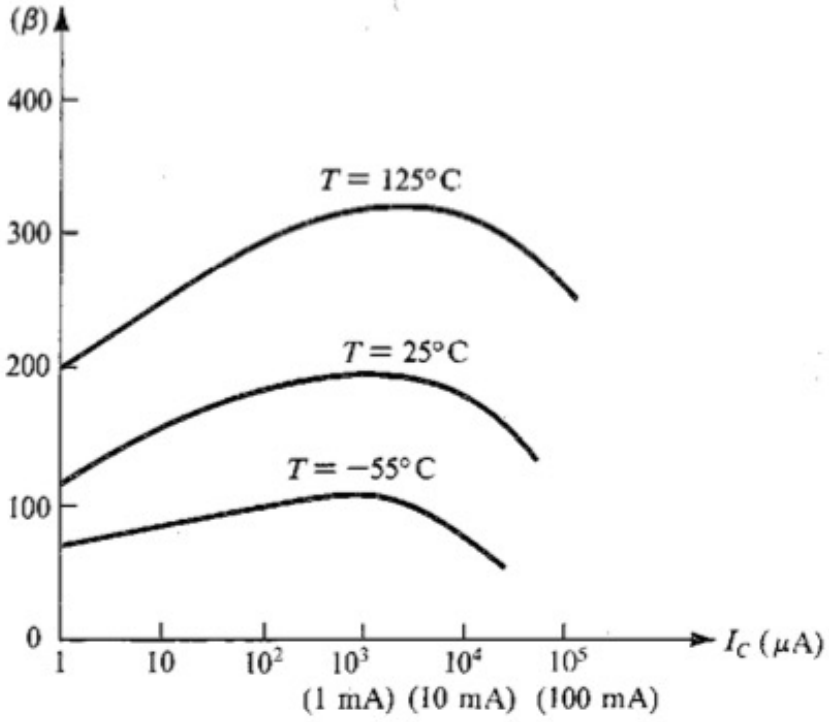
IV Characteristics

The current in the collector is a result of a forward-biased pn junction (EB junction)



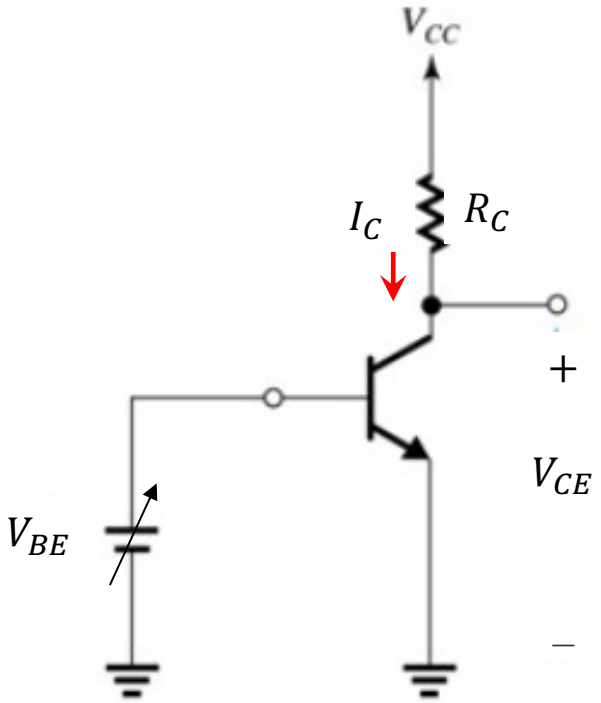
IV Characteristics

The amplification parameter of a BJT β is a function of many things; most importantly temperature.

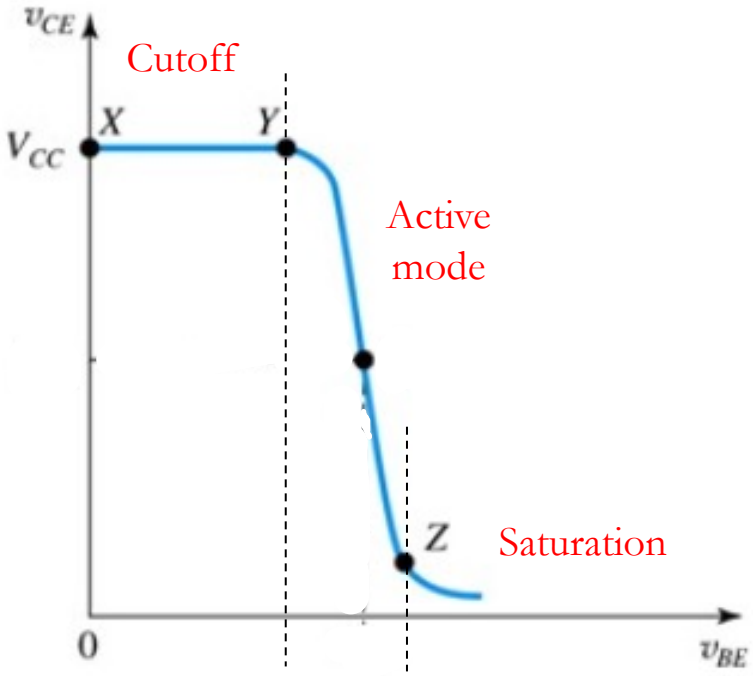


The BJT Transfer Function

The transfer function indicated the region of operation of each mode. It also help identify the possible applications.



$$V_{CE} = V_{CC} - I_C R_C$$

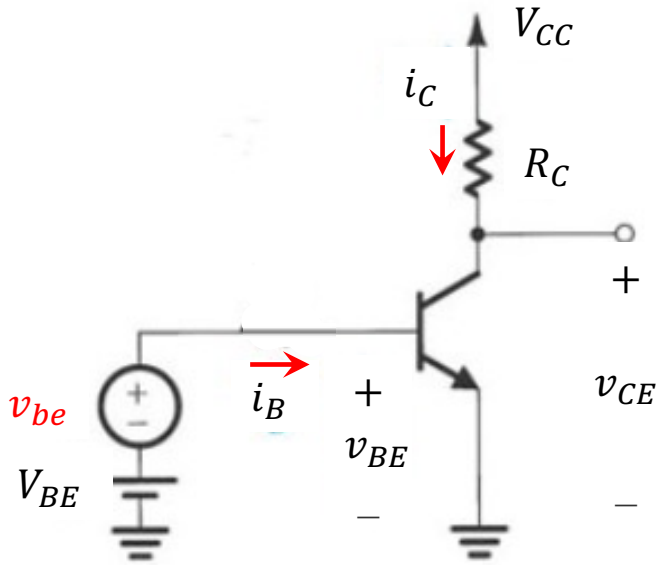


Bipolar Junction Transistors

- Small Signal Analysis.
- The BJT as an Amplifier.

Small Signal Analysis of BJTs

Let's consider the circuit below, where both DC and ac signals (large signal) are applied. We try to write the circuit variables as two components: DC and ac.



$$v_{BE} = V_{BE} + v_{be}$$

$$i_C = I_s\, e^{v_{BE}/V_T} = I_s\, e^{V_{BE}/V_T} e^{v_{be}/V_T}$$

$$i_C = I_C\, e^{v_{be}/V_T}$$

Small-signal approximation: $v_{be} \ll V_T$

$$i_C = I_C\,(1 + \frac{v_{be}}{V_T})$$

$$i_C = I_C + \frac{I_C}{V_T} v_{be}$$

DC component

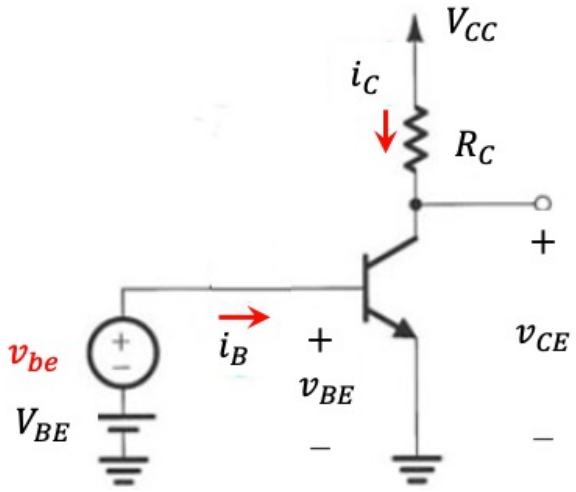
ac component

$$i_c = \frac{I_C}{V_T} v_{be} = g_m v_{be}$$

g_m Transconductance (A/V)

Small Signal Analysis

Let's consider the circuit below, where both DC and ac signals (large signal) are applied. We try to write the circuit variables as two components: DC and ac.



$$i_C = I_C + \frac{I_C}{V_T} v_{be} \qquad i_c = g_m v_{be}$$

$$i_B = \frac{I_C}{\beta} + \frac{I_C}{\beta V_T} v_{be} \qquad i_b = \frac{g_m}{\beta} v_{be}$$

$$i_E = \frac{I_C}{\alpha} + \frac{I_C}{\alpha V_T} v_{be} \qquad i_e = \frac{g_m}{\alpha} v_{be}$$

$g_m = \frac{I_C}{V_T}$ (A/V)
Transconductance

$r_\pi = \frac{\beta}{g_m}$ (Ω)
Base resistance

$r_e = \frac{\alpha}{g_m}$ (Ω)
Emitter resistance

$r_o = \frac{V_A}{I_C}$ (Ω)
Output resistance

Note that r_π and r_e both represent the base-emitter circuit; one looking from the base side (r_π) and the other looking from the emitter side (r_e). Also,

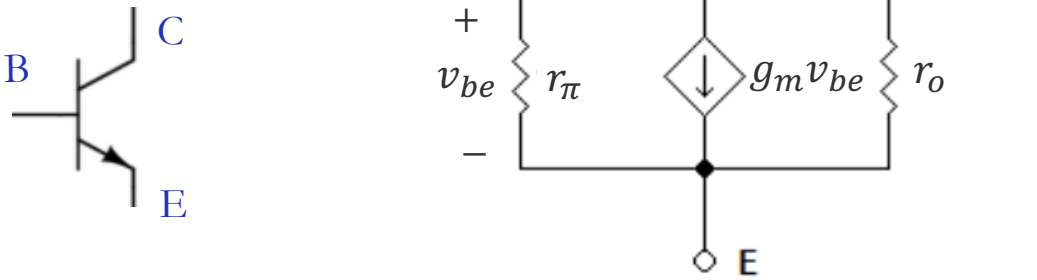
$$r_\pi = (\beta + 1)\, r_e$$

Based on these equations, we can develop a small-signal equivalent circuit of the BJT transistor.

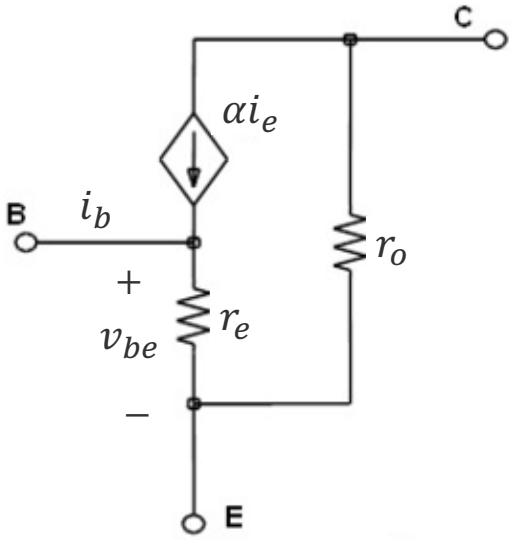
BJT Small-signal Equivalent Circuit

Based on these equations, we can develop a small-signal equivalent circuit of the BJT transistor.

The pi-model: involves the base resistance



The T-model: involves the emitter resistance



Small-signal parameters of the BJT

$g_m = \frac{I_C}{V_T}$ (A/V)
Transconductance

$r_\pi = \frac{\beta}{g_m}$ (Ω)
Base resistance

$r_e = \frac{\alpha}{g_m}$ (Ω)
Emitter resistance

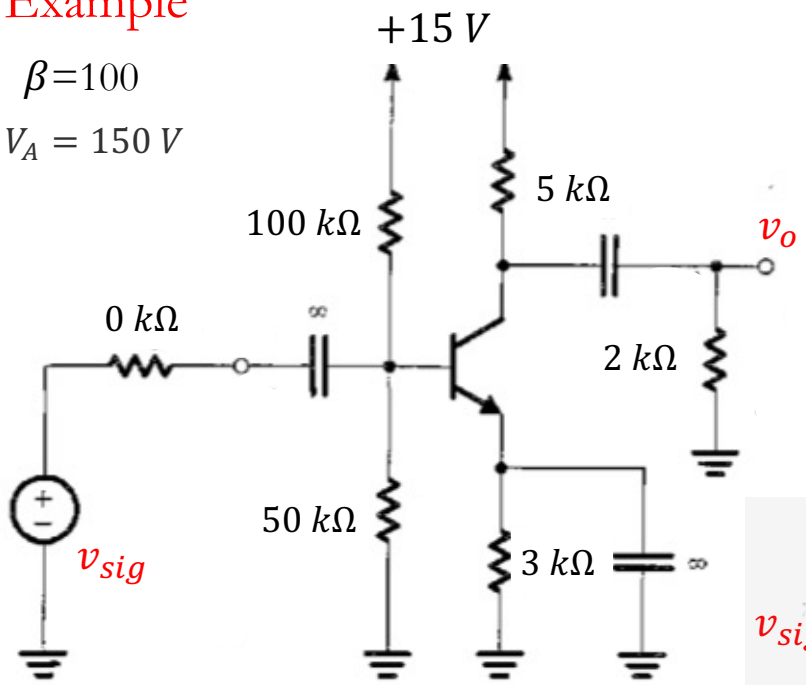
$r_o = \frac{V_A}{I_C}$ (Ω)
Output resistance

Small Signal Analysis

Example

β=100

$V_A = 150\ V$



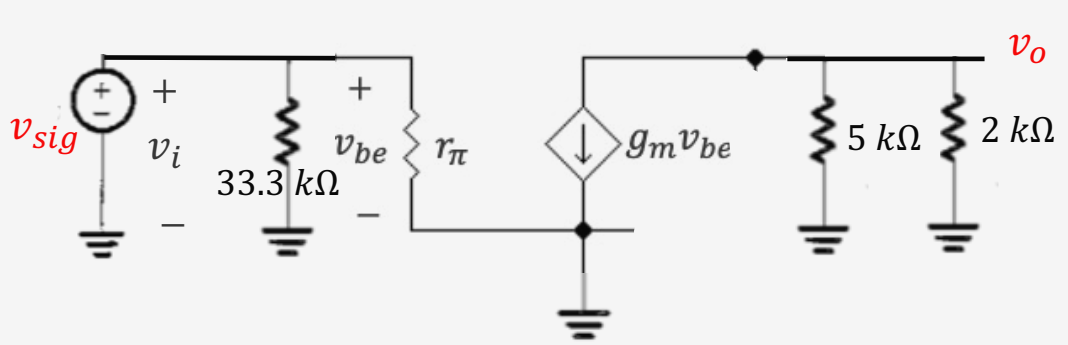
DC Analysis: Refer to week 11 examples

$$I_C = 1.28\ mA$$

$$g_m = \frac{I_C}{V_T} = 50\ \text{mA/V} \qquad r_\pi = \frac{\beta}{g_m} = 2\ k\Omega$$

$$r_e = \frac{\alpha}{g_m} = 19.8\ \Omega \qquad r_o = \frac{V_A}{I_C} = 117\ k\Omega$$

AC Analysis: Use pi-model, ignore r_o



$$v_o = -g_m v_{be}(5 \parallel 2)k$$

$$v_i = v_{be}$$

$$A_v = \frac{v_o}{v_i} = -g_m(5 \parallel 2)k = -71.4\ (V/V)$$

$$R_i = (33.3 \parallel r_\pi)k = 1.88\ k\Omega$$

$$R_o = 5\ k\Omega$$

Q & A

Nano Photonics and Plasmonics Research

Send me an email if
you have any
questions.